FD Closure

Closures – all logically implied FDs

Let F be a set of FDs. Let A, B be sets of attributes.

The closure of A (A^+) = the set of attributes B such that $A \rightarrow B$.

The closure of F (F^+) = the set of FDs logically implied by F.

A^+ = the set of attributes B such that $A \rightarrow B$.

Closure of attribute sets

Given:

{name} $\rightarrow$ {color}

{category} $\rightarrow$ {dept}

{color, category} $\rightarrow$ {price}

Some closures:

{name}$^+$	= {name, color}
{name, category}$^+$	= {name, category, color, dept, price}
{color}$^+$	= {color}

A^+ = the set of attributes B such that $A \rightarrow B$.

Algorithm for closure of attribute sets

$Closure_F(A)$ =

$Result = A$

Repeat until $Result$ doesn't change:

For each FD $B \rightarrow C$ in F:

if $B \subseteq Result$,

then add C to $Result$

Return $Result$

Given:

{name} → {color}

{category} → {dept}

{color, category} → {price}

Compute:

{name, category}$^+$

A^+ = the set of attributes B such that $A \rightarrow B$.

Activity – use closure algorithm

$Closure_F(A)$ =

$Result = A$

Repeat until $Result$ doesn't change:

For each FD $B \rightarrow C$ in F:

if $B \subseteq Result$,

then add C to $Result$

Return $Result$

Given:

{a, b} → {c}

{a, d} → {e}

{b} → {d}

{a, f} → {b}

Compute:

{a, b}$^+$ =

{a, f}$^+$ =

F^+ = the set of FDs logically implied by F.

Closure of FD set

$Closure(F)$ =

$Result = \emptyset$

For each subset A of attributes used in F:

~~For each $A \rightarrow B$ provable:~~

For each $B \subseteq A^+$:

Add $A \rightarrow B$ to $Result$.

Return $Result$.

F^+ = the set of FDs logically implied by F.

Closure of FD set – example

Given:

{a, b} → {c}
{a, d} → {b}
{b} → {d}

Compute attribute set closures:

{a}$^+$ = {a}
{b}$^+$ = {b, d}
{c}$^+$ = {c}
{d}$^+$ = {d}
{a, b}$^+$ = {a, b, c, d}
{a, c}$^+$ = {a, c}
{a, d}$^+$ = {a, b, c, d}
{b, c}$^+$ = {b, c, d}
{b, d}$^+$ = {b, d}
{c, d}$^+$ = {c, d}
{a, b, c}$^+$ = {a, b, c, d}
{a, b, d}$^+$ = {a, b, c, d}
{a, c, d}$^+$ = {a, b, c, d}
{b, c, d}$^+$ = {b, c, d}
{a, b, c, d}$^+$ = {a, b, c, d}

Compute FDs:

{a} → {a}
{b} → {b}, … → {d}, … → {b, d}
{c} → {c}
{d} → {d}
{a, b} → {a, b, c, d}
{a, c} → {a, c}
{a, d} → {a, b, c, d}
{b, c} → {b, c, d}
{b, d} → {b, d}
{c, d} → {c, d}
{a, b, c} → {a, b, c, d}
{a, b, d} → {a, b, c, d}
{a, c, d} → {a, b, c, d}
{b, c, d} → {b, c, d}
{a, b, c, d} → {a, b, c, d}

If we take $A \rightarrow B$ to be shorthand for $A \rightarrow B'$, where $B' \subseteq B$.

F^+ = the set of FDs logically implied by F.

Closure of FD set – example

Given:

{a, b} → {c}
{a, d} → {b}
{b} → {d}

…

Compute FDs:

{a} → {a}
{b} → {b}, … → {d}, … → {b, d}
{c} → {c}
{d} → {d}
{a, b} → {a, b, c, d}
{a, c} → {a, c}
{a, d} → {a, b, c, d}
{b, c} → {b, c, d}
{b, d} → {b, d}
{c, d} → {c, d}
{a, b, c} → {a, b, c, d}
{a, b, d} → {a, b, c, d}
{a, c, d} → {a, b, c, d}
{b, c, d} → {b, c, d}
{a, b, c, d} → {a, b, c, d}

Shorthand version:

{b} → {d}

{a, b} → {c, d}

{a, d} → {b, c}
{b, c} → {d}

{a, b, c} → {d}
{a, b, d} → {c}
{a, c, d} → {b}

Eliminating trivial FDs $A \to B$, where $B \subseteq A$.

Replacing FDs $A \to AB$ with $A \to B$.

FD covers & equivalence

F covers G iff G can be inferred from F. I.e., $G^+ \subseteq F^+$.

F and G are equivalent iff $F^+ = G^+$.

FD covers & equivalence

F covers G iff G can be inferred from F. I.e., $G^+ \subseteq F^+$.

Given:
{a, b} → {c}
{a, d} → {b}
{b} → {d}

closure

Compute FDs:
{a} → {a}
{b} → {b, d}
{c} → {c}
{d} → {d}
{a, b} → {a, b, c, d}
{a, c} → {a, c}
{a, d} → {a, b, c, d}
{b, c} → {b, c, d}
{b, d} → {b, d}
{c, d} → {c, d}
{a, b, c} → {a, b, c, d}
{a, b, d} → {a, b, c, d}
{a, c, d} → {a, b, c, d}
{b, c, d} → {b, c, d}
{a, b, c, d} → {a, b, c, d}

covered by

Shorthand version:

{b} → {d}

{a, b} → {c, d}

{a, d} → {b, c}
{b, c} → {d}

{a, b, c} → {d}
{a, b, d} → {c}
{a, c, d} → {b}

SUPERKEYS & KEYS

Superkeys & keys

Intuition: R's attributes should be dependent on exactly R's primary key.

A set A of attributes is a superkey of relation R iff $A \rightarrow R$.

Equivalently, iff $A^+ = R$.

A (candidate) key is a minimal superkey.

We pick a key as primary key.

Activity: Find superkeys & keys

Product(name, price, category, color)

{name, category} → {price}

{category} → {color}